Sreekaran Reddy Ramasahayam

(913) 299-7216 | sreekaran.2021@gmail.com | linkedin.com/in/sreekaranreddy1704 | github.com/Sreekaran1704

Skills

Machine Learning: Regression, Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Data Processing: Python (Pandas, NumPy, scikit-learn), SQL, Data Cleaning, EDA

MLOps: MLflow, DVC, Docker, FastAPI, CI/CD (GitHub Actions), Model Versioning

GenAI & NLP: Pytorch, LLM Fine-tuning, LORA, PEFT, Prompt Engineering, Evaluation Pipelines

Cloud: AWS (EC2, S3, ECR), PostgreSQL, MongoDB, Git, Linux, Stremlit

Education

University of Missouri-KansasCity — **Aug 2024 - May 2026**

Masters in Computer Science — *GPA: 3.96*

- Relevant coursework: Generative AI, Principles of Data Science, Statistical Learning, Cloud Computing.

Work Experience

Sree Nirman — **May 2023 – Apr 2024**

Machine Learning Engineer Intern (Analytics & MLOps) — *Hyderabad, India*

- Started with raw inconsistent construction data (50k+ records), so I built an end-to-end pipeline using Python to handle data cleaning, validation, and feature engineering, creating reliable, model-ready datasets reducing data preparation time by 20%.
- To support more accurate cost estimation, developed regression-based machine learning models on historical construction data, capturing patterns across key project attributes to improve planning and budgeting.
- To improve model performance and ensure consistency, experimented with multiple approaches and tracked results using MLflow, selecting models that produced more stable and usable cost estimates for decision-making.
- To make the solution production-ready, deployed the model as a containerized FastAPI service on AWS EC2 using Docker, and implemented CI/CD pipelines with GitHub Actions to enable reliable deployments and consistent prediction outputs.

Avanthi High School — **Apr 2022 – Jan 2023**

Data Analyst Intern — *Warangal, India*

- Started with analyzing 12K+ student academic records where fee concessions were manually assigned, so by using Python (Pandas, NumPy) I handled the missing values, inconsistencies, and outliers to improve data quality and consistency by 30%.
- To understand inefficiencies in concession allocation, analyzed department-wise financial data and historical student records, identifying recurring anomalies and patterns that highlighted 10–15% potential cost savings.
- To make concession decisions more objective, built an XGBoost-based machine learning model to predict student performance (future exam outcomes) using academic history and entrance scores, mapping predictions to standardized concession brackets.
- Fine-tuned a GPT-2 model as an LLM-based evaluator using historical student responses (5K+ samples) and scoring labels, and integrated it with ML predictions using a weighted approach improving evaluation consistency and reducing subjective bias.
- Replaced manual evaluation with a reusable data and modeling pipeline, documenting preprocessing workflows and reducing turnaround time by 25%, while improving consistency and reducing bias in concession allocation decisions.

Assistantship

University of Missouri-KansasCity — **Aug 2025 – Present**

Information Services Lab Assistant — *KansasCity, Missouri*

- Mentoring undergraduate students on data analysis workflows, guiding them through statistics and EDA while improving the quality, consistency, and reproducibility of their analytical work.
- Reviewing and resolving reproducibility issues while maintaining lab system documentation to improve it's accuracy and consistency.

Projects

Genre-Controlled Story Generation using LoRA (Gemma Fine-Tuning) — PyTorch | PEFT (QLoRA) | Transformers

- Fine-tuned Gemma 3–1B with QLoRA/PEFT for controlled text generation, building a reproducible instruction-tuning pipeline with structured prompt formatting, dataset curation, and held-out evaluation..
- Evaluated base vs adapted models using validation loss, perplexity, genre fidelity, coherence, and LLM-as-judge scoring to analyze whether fine-tuning improved controllability and output quality..
- Experimented with decoding parameters such as temperature and top-k to improve generation quality while tracking failure cases including repetition, weak endings, and genre drift.
- Developed a Streamlit web app enabling users to input prompts, select target genres, and generate stories using the fine-tuned Gemma model with adjustable decoding parameters including temperature, top-k, and top-p for controllable text generation.

Vehicle Insurance Eligibility Prediction & MLOps Pipeline — Python | MLflow | DVC | AWS

- Built an end-to-end pipeline for insurance eligibility prediction, handling data ingestion, preprocessing, feature engineering, and model training on structured datasets.
- Deployed the model as a FastAPI service on AWS EC2 with CI/CD via GitHub Actions, ensuring reproducible experiments and scalable deployment.